

Phase 8: Data Management & Deployment

AgriConnect

Salesforce-Based Farmer Support & Marketplace System

For AgriConnect – Salesforce-Based Farmer Support & Marketplace System, proper data management and smooth deployment are essential to ensure reliability, accuracy, and scalability. Salesforce provides a variety of tools and methods for handling data and moving metadata across environments.

Step 1: Data Import Wizard

- The Data Import Wizard is a point-and-click tool inside Salesforce for importing records up to 50,000 at a time. It is ideal for adding farmers, crop listings, marketplace entries, and orders.
- Steps:
 - Go to Salesforce → Setup → Quick Find → Data Import Wizard → Launch Wizard.
 - Select the object (e.g., CropListing__c, Orders__c, Farmers__c).
 - Choose the action: Add new records or Update existing records using External ID.
 - Map CSV fields to Salesforce fields.
 - Click Start Import.
- Image Placement:
 - Screenshot of the Data Import Wizard page with object selection.
 - Screenshot of the field mapping screen.

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List View Button Layout

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Scoping Rules

Order Custom Field

External Id

Back to Order

Validation Rules (0)

Custom Field Definition Detail

Edit

Set Field-Level Security

View Field Accessibility

Where is this used?

Field information

Field Label

External Id

Field Name

External Id

API Name

EXTERNAL_ID__c

Description

Field Type

Object Name

Object Name

Date Type

Object

Data Source

Field Usage

Data Security Level

Compliance Categorization

Created By

SAL RISHMA MUKKAMALLA

9/29/2020, 7:32 AM

Modified By

SAL RISHMA MUKKAMALLA

9/29/2020, 7:32 AM

General Options

Required

Not Set

Case Sensitive

External ID

Default Value

Text Options

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Almost done

Choose data

Edit mapping

Start import

Edit Field Mapping: Consultations

Your file has been auto-mapped to existing Salesforce fields, but you can edit the mappings if you wish. Unmapped fields will not be imported.

Help for this page

Edit	Mapped Salesforce Object	CSV Header	Example	Example	Example
Map	Unmapped @	External_Id__c	CNS-001	CNS-002	CNS-003
Change	Consultation Name	Name	Consult 1	Consult 2	Consult 3
Map	Unmapped @	Farmer__r.External_Id__c	FRM-001	FRM-002	FRM-003
Map	Unmapped @	Consultant__r.External_Id__c	USR-001	USR-002	USR-003
Change	Topic__c	Topic__c	Pest control	Soil nutrition	Irrigation
Map	Unmapped @	Consultation_Date__c	05-09-2025	06-09-2025	07-09-2025
Change	Notes__c	Notes__c	Checked for pest	Recommended	Advised watering schedule

CancelPreviousNext

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Almost done

Choose data

Edit mapping

Start import

Edit Field Mapping: Crop Listings

Your file has been auto-mapped to existing Salesforce fields, but you can edit the mappings if you wish. Unmapped fields will not be imported.

Help for this page

Edit	Mapped Salesforce Object	CSV Header	Example	Example	Example
Map	Unmapped @	External_Id__c	CL-001	CL-002	CL-003
Change	Crop Listing Name	Name	Wheat Batch A	Rice Batch B	Maize Batch C
Map	Unmapped @	Crop_Type__c	Wheat	Rice	Maize
Map	Unmapped @	Variety__c	Durum	Basmati	Yellow
Change	Quantity__c	Quantity__c	1000	500	800
Change	Unit__c	Unit__c	kg	kg	kg
Change	Price__c	Price__c	12000	8000	10000

CancelPreviousNext

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Almost done

Choose data

Edit mapping

Start import

Edit Field Mapping: Crop Listings

Your file has been auto-mapped to existing Salesforce fields, but you can edit the mappings if you wish. Unmapped fields will not be imported.

Help for this page

Edit	Mapped Salesforce Object	CSV Header	Example	Example	Example
Change	External Id	External_Id__c	CL-001	CL-002	CL-003
Change	Crop Listing Name	Name	Wheat Batch A	Rice Batch B	Maize Batch C
Change	Crop Type	Crop_Type__c	Wheat	Rice	Maize
Change	Variety	Variety__c	Durum	Basmati	Yellow
Change	Quantity__c	Quantity__c	1000	500	800
Change	Unit__c	Unit__c	kg	kg	kg
Change	Price__c	Price__c	12000	8000	10000

CancelPreviousNext

Setup

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Search Setup

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Setup > OBJECT MANAGER

Almost done

Choose data

Edit mapping

Start import

Review & Start Import

Review your import information and click Start Import.

Help for this page

Your selections:

Crop Listings

Add new and update existing records

croplisting.csv

Your import will include:

Mapped fields

10

Your import will not include:

Unmapped fields

0

CancelPreviousStart Import

Step 2: Data Loader

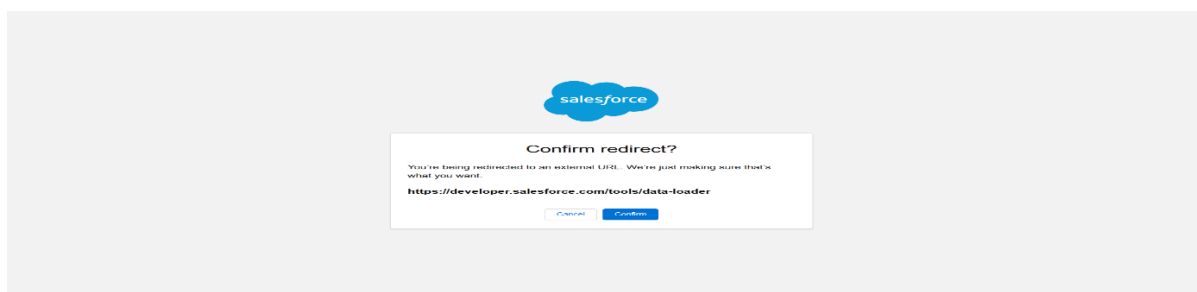
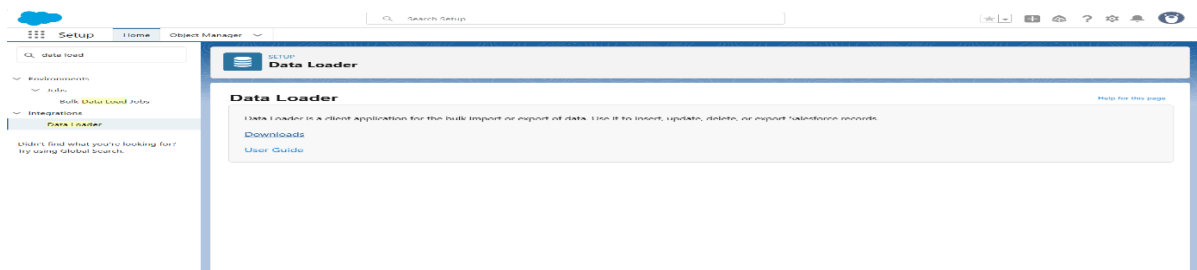
- Data Loader is a **client application** used for **bulk data operations** (insert, update, upsert, delete, export). It is recommended for large datasets exceeding 50,000 records.

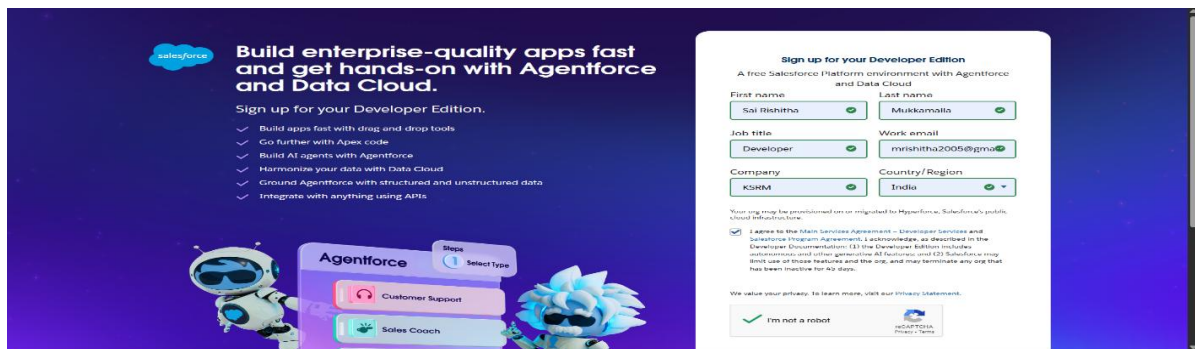
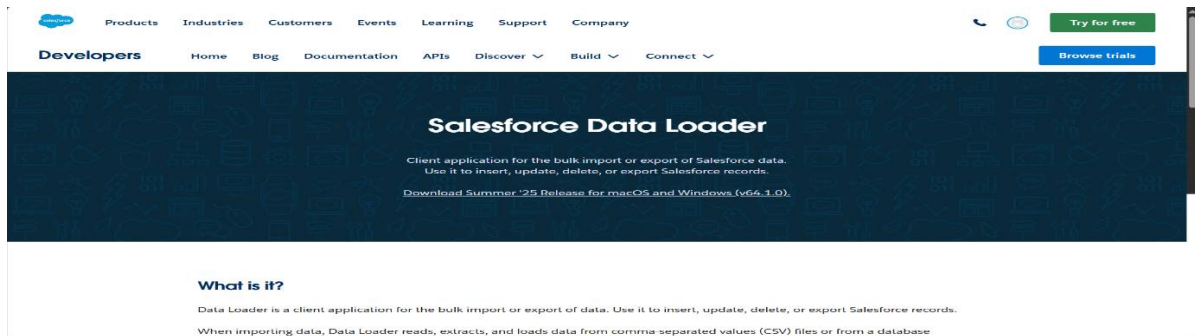
- **Steps:**

- Download and install Data Loader from Salesforce.
- Open the application and login with your Salesforce credentials.
- Choose the operation: **Insert, Update, Upsert, Delete, Export**.
- Select the target object (e.g., CropListing__c).
- Upload the CSV file and map fields to Salesforce.
- Run the operation and download success/error files.

- **Image Placement:**

- Screenshot of Data Loader login screen.
- Screenshot showing operation selection (Insert/Update/Upsert).
- Screenshot of field mapping window.



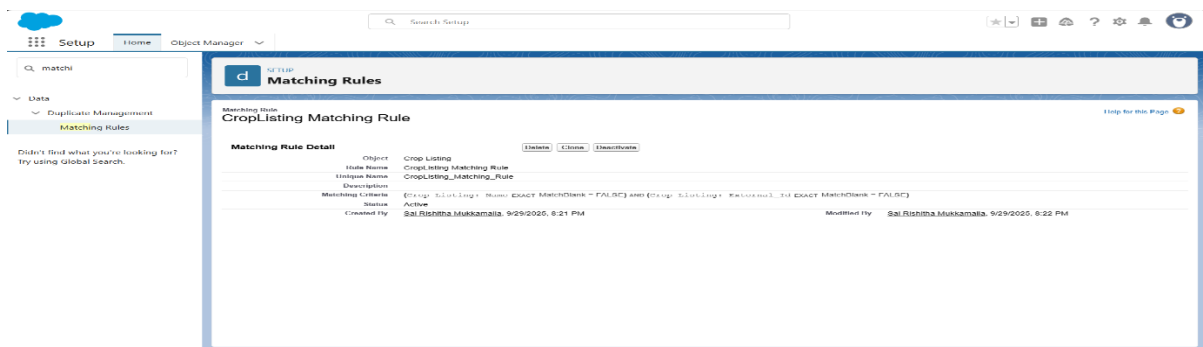
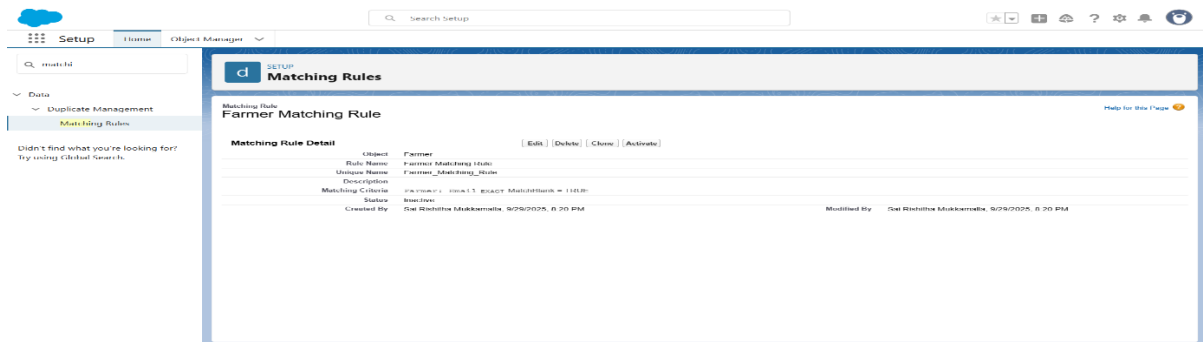


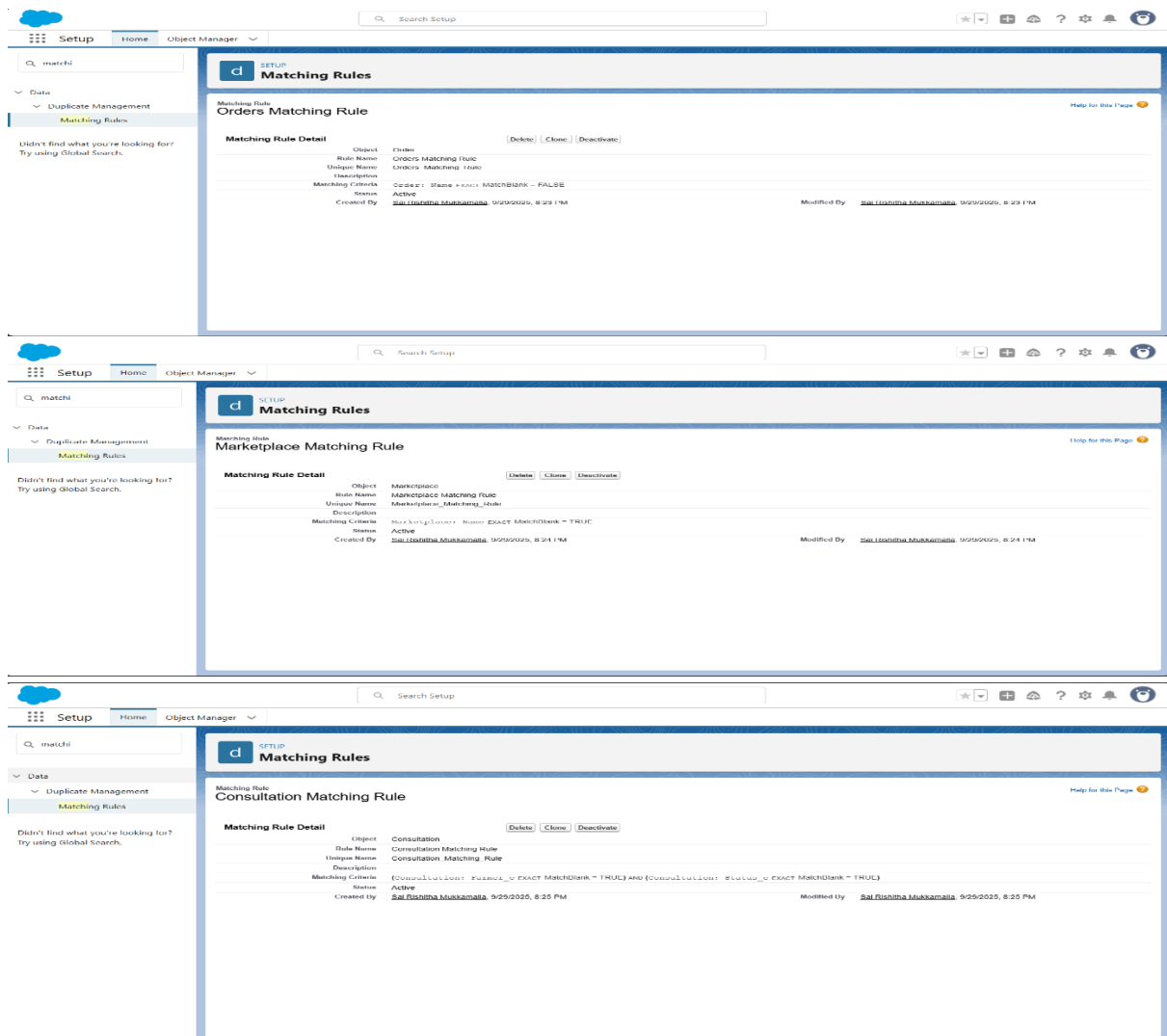
Step 3: Duplicate Rules

➤ Create Matching Rules

- Matching rules define **how Salesforce identifies duplicates**.
- **Where:**
 - Setup → Quick Find → **Duplicate Management** → **Matching Rules** → New Rule
- **Objects & Matching Logic:**
 - **Farmers__c**
 - Match by: **Email, Phone Number, Aadhaar Number**
 - Logic: Exact or fuzzy match to prevent duplicate farmer entries
 - **CropListing__c**
 - Match by: **Crop Name, Crop Type, Farmer__c**
 - Ensures no duplicate crop entries for the same farmer
 - **Orders__c**
 - Match by: **Order Number, Farmer__c**

- Prevents duplicate orders for the same farmer
- **Marketplace__c**
 - Match by: **Listing Name, Crop__c, Farmer__c**
 - Avoids duplicate marketplace listings
- **Consultation__c**
 - Match by: **Farmer__c, Date**
 - Prevents duplicate consultations for the same day
- **Knowledge__c**
 - Match by: **Title**
 - Avoids duplicate knowledge articles
- **Action after creation:**
 - Click **Save** → Activate the rule.
- **Image Placement:**
 - Screenshot of Matching Rules list.
 - Screenshot showing field criteria for one example object (e.g., Farmers__c).

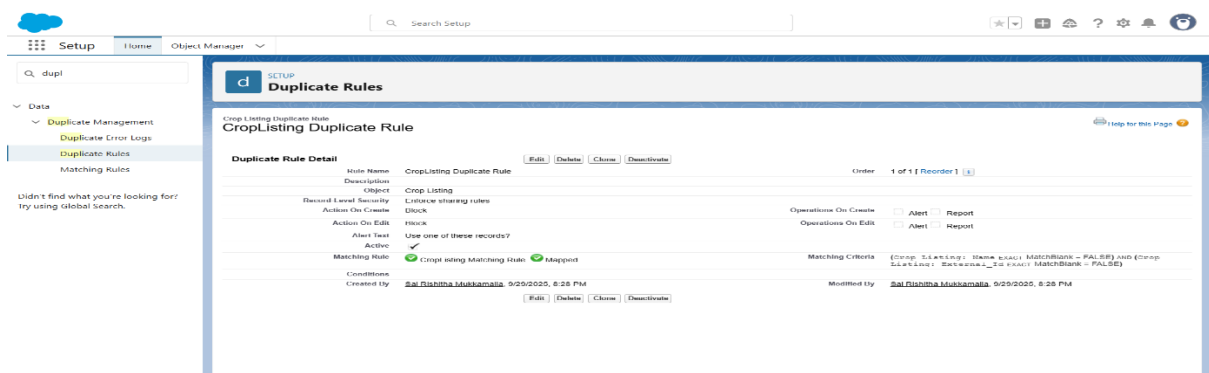
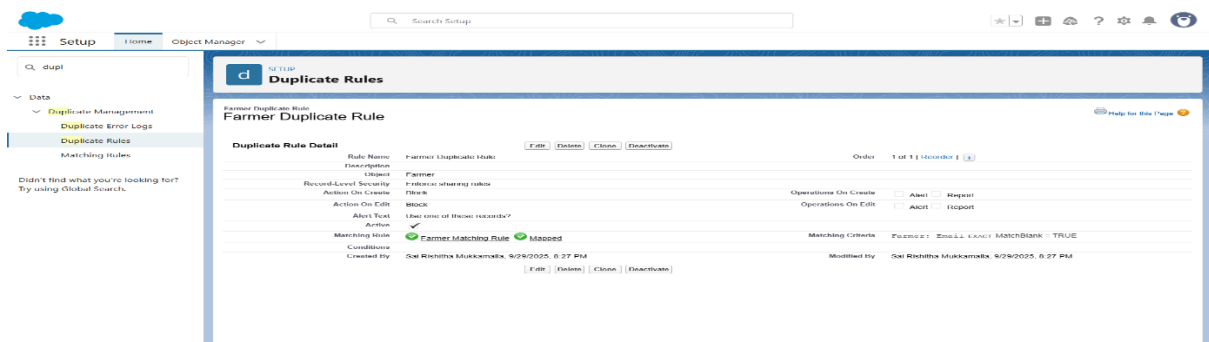


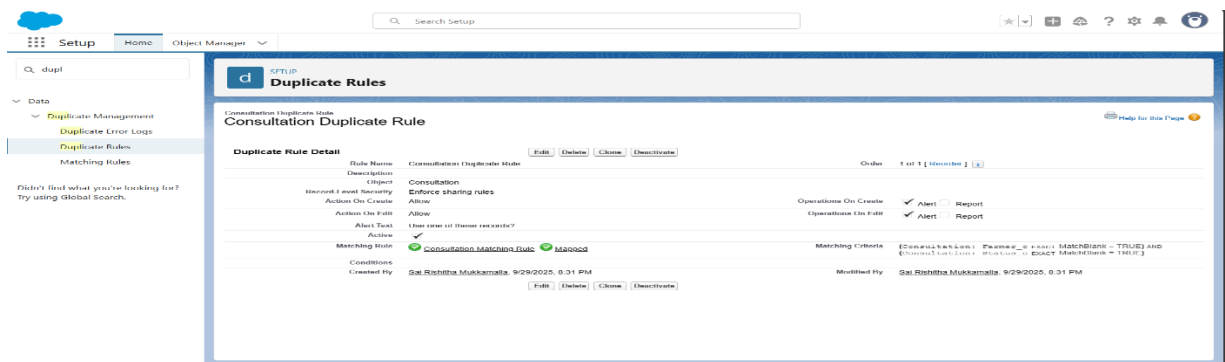
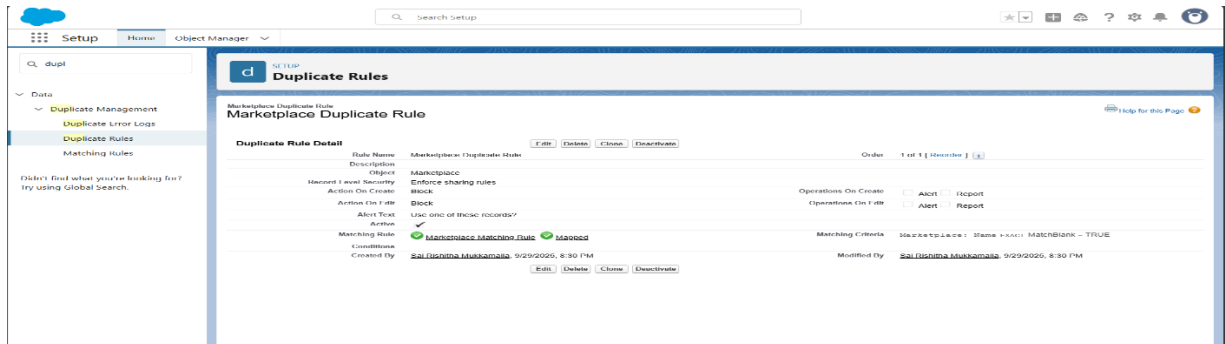
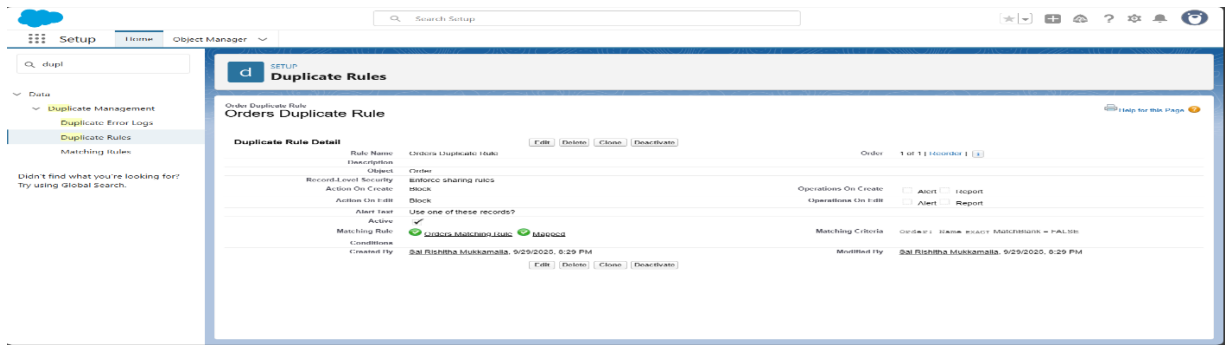


➤ Create Duplicate Rules

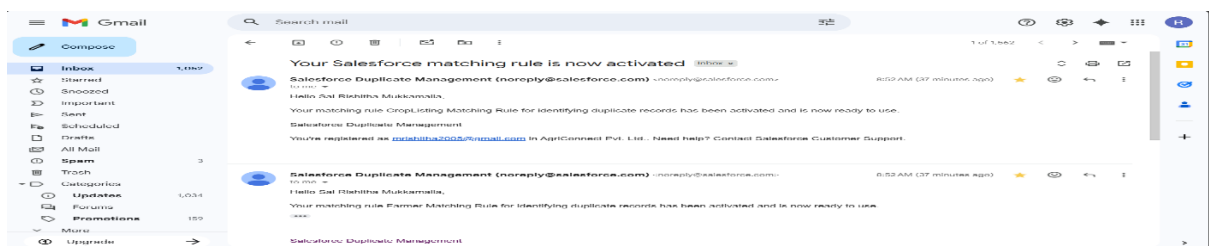
- Duplicate Rules determine **what happens when a duplicate is detected.**
- **Where:**
 - Setup → Quick Find → **Duplicate Management** → **Duplicate Rules** → New Rule
- **Steps for each object:**
 - Select the object (e.g., Farmers__c).
 - Name the rule: Farmers Duplicate Rule
 - Apply the **matching rule** you created in Step 1.
 - Choose **Action on Create/Edit:**
 - **Block** → Prevent saving duplicates

- **Alert** → Warn user but allow save
- **Report** → Log duplicates for review
- Decide **Record-Level Security** (who can see alerts).
- Activate the rule.
- **Objects & Rules Applied:**
 - **Farmers__c** → Block or Alert on duplicate Email/Aadhaar
 - **CropListing__c** → Block duplicate crop listings
 - **Orders__c** → Alert if order number exists for the same farmer
 - **Marketplace__c** → Block duplicate listings
 - **Consultation__c** → Alert if same farmer has consultation on the same date
 - **Knowledge__c** → Block duplicate titles
- **Image Placement:**
 - Screenshot of Duplicate Rules list.
 - Screenshot of rule setup page showing the action (Block/Alert).





- **Test Duplicate Rules**
 - Try creating a duplicate record for each object.
 - Confirm the system either **blocks, alerts, or reports** duplicates as configured.
- **Image Placement:**
 - Screenshot showing duplicate alert message.



- **Activate and Maintain**

- Ensure all rules are **activated**.
- Periodically review **matching criteria** and adjust for new business rules.
- Helps AgriConnect maintain **clean, accurate data** for reporting and automation.

Step 4: Data Export & Backup

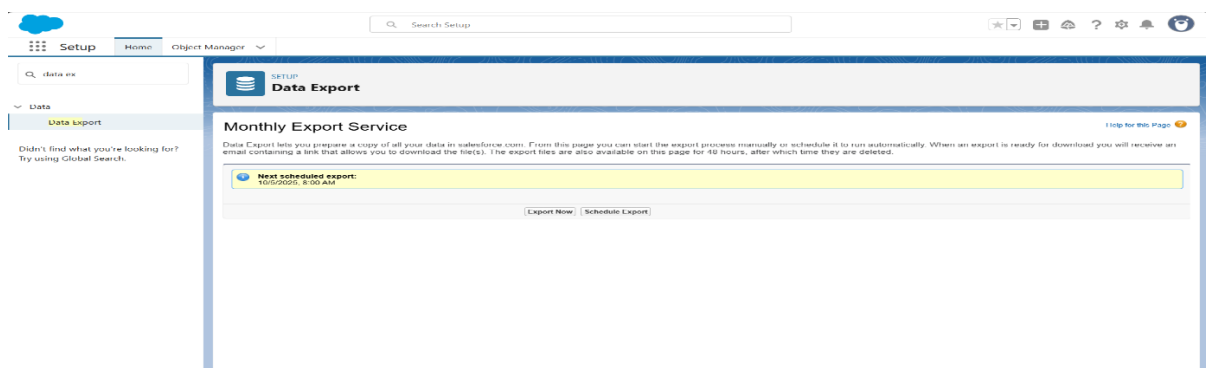
➤ Regularly backup Salesforce data to CSV to prevent data loss and support recovery.

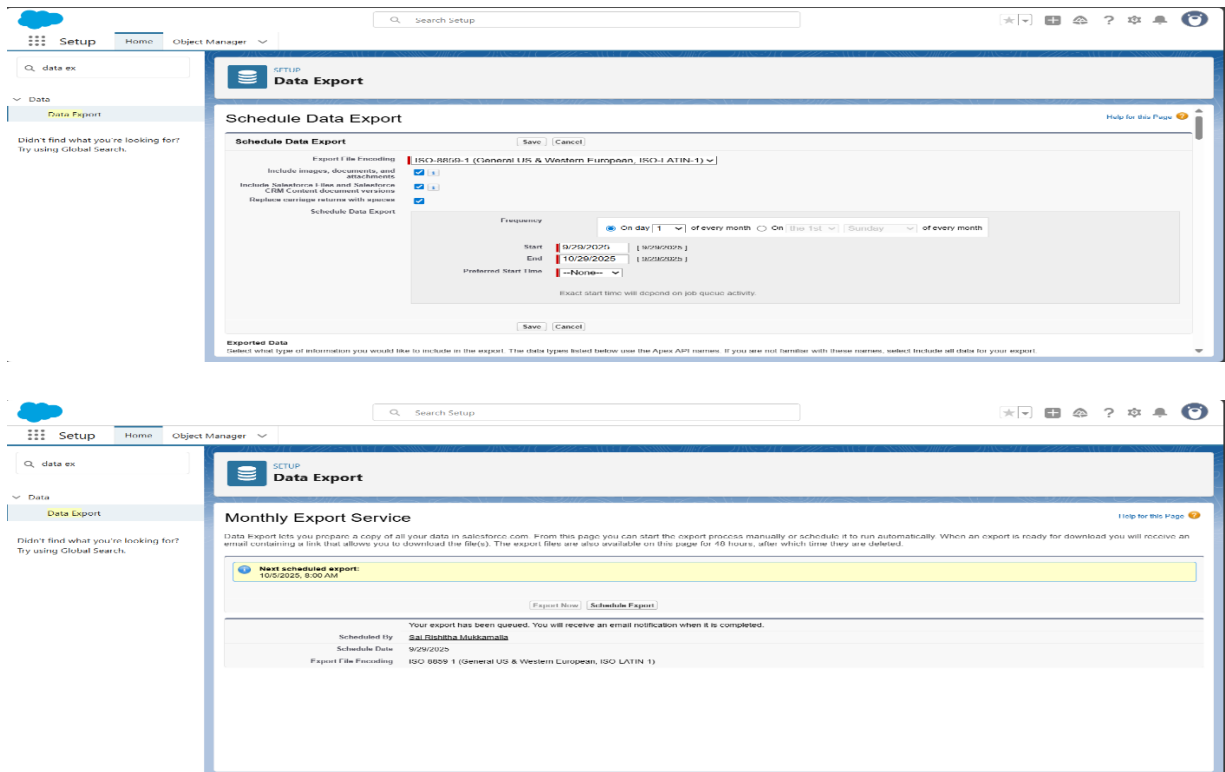
➤ **Steps:**

- Go to **Salesforce → Setup → Quick Find → Data Export**.
- Choose **Export Now** (one-time) or **Schedule Export** (weekly/monthly).
- Select the objects to export:
 - CropListing__c, Orders__c, Farmers__c, Marketplace__c, Consultation__c, Knowledge__c.
- Optional: Include attachments and documents.
- Click **Start Export** → Salesforce generates ZIP files with CSVs.
- Download and store them securely.

➤ **Image Placement:**

- Screenshot of **Data Export** page showing object selection.
- Screenshot showing **Start Export** button and progress.





Step 5: Change Sets

- Deploy Salesforce metadata (objects, fields, Apex, LWCs, Flows) from **Sandbox → Production**.

➤ Steps:

- Go to **Setup → Quick Find → Change Sets → Outbound Change Sets**.
- Click **New** → Enter **Change Set Name** and **Description**.
- Click **Add Components** → Select component types:
 - Custom Objects: CropListing__c, Orders__c, Farmers__c, Marketplace__c, Consultation__c, Knowledge__c
 - Fields, Apex Classes, LWCs, Flows, Validation Rules
- Click **Upload** → Select **Target Org (Production)**.
- In Production → Setup → **Inbound Change Sets** → Review → Click **Deploy**.

➤ Image Placement:

- Screenshot of **Outbound Change Set** page.

- Screenshot showing **Add Components** interface.

Step 6: Unmanaged vs Managed Packages

- Bundle and deploy Salesforce components as a **package**.
- **Steps:**
 - Go to **Setup** → **Quick Find** → **Packages** → **New**.
 - Enter:
 - Package Name: AgriConnect Package
 - Description: All AgriConnect metadata
 - Choose **Package Type**:
 - **Unmanaged**: Editable in target org (for internal deployment)
 - **Managed**: Versioned, restricted (for AppExchange)
 - Click **Add Components** → Include:
 - Custom Objects, Fields, Apex Classes, LWCs, Flows, Validation Rules
 - Upload/Install in target org.
- **Image Placement:**
 - Screenshot of **Packages** page.
 - Screenshot showing **Add Components** window.

Step 7: ANT Migration Tool

- Scripted metadata deployment using Salesforce Metadata API. Useful for **automated CI/CD or offline deployments**.
- **Steps:**
 - Download **ANT Migration Tool** from Salesforce.
 - Configure build.xml and build.properties with org credentials.
 - Use commands:
 - ant retrieve → Download metadata
 - ant deploy → Push metadata to target org
 - Verify logs for success/error.

➤ **Image Placement:**

- Screenshot of **ANT folder structure**.
- Screenshot showing **command line retrieve/deploy logs**.

Step 8: VS Code & SFDX

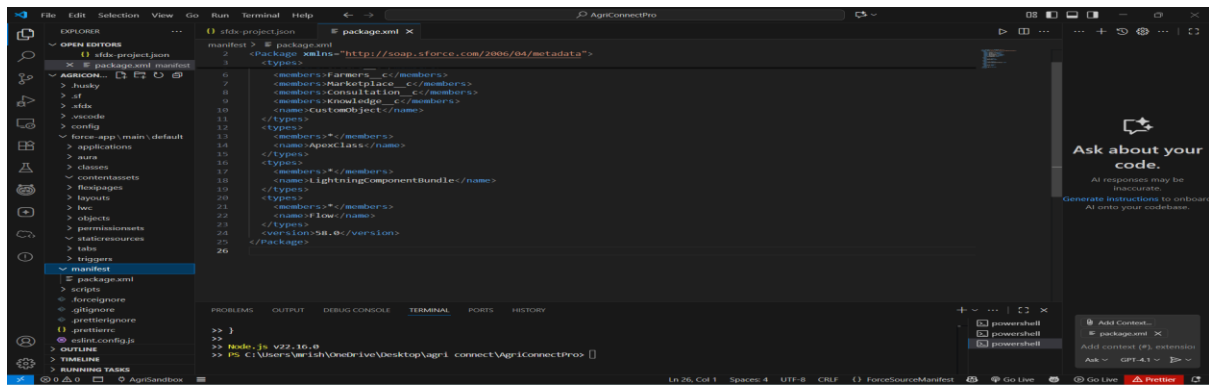
➤ Modern **source-driven development environment** for Salesforce.
Supports version control, local development, and metadata deployment.

➤ **Steps:**

- Install **VS Code + Salesforce Extensions Pack**.
- Authorize org:
`sfdx force:auth:web:login -a AgriSandbox`
- Create **manifest/package.xml** to define metadata.
- Retrieve metadata:
`sfdx force:source:retrieve -x manifest/package.xml -u AgriSandbox`
- Make changes locally (objects, fields, Apex, LWCs, flows).
- Deploy changes:
`sfdx force:source:deploy -p force-app`
- Validate deployment (dry run):
`sfdx force:source:deploy -x manifest/package.xml --checkonly -u AgriSandbox`
- Version control:
`git init`
`git add .`
`git commit -m "AgriConnect initial version"`

➤ **Image Placement:**

- Screenshot of **VS Code with Salesforce Extensions**.
- Screenshot of **terminal showing retrieve/deploy commands**.
- Screenshot of **folder structure** (objects/, classes/, lwc/, flows/).



Conclusion

- Phase 8 ensures that AgriConnect's Salesforce environment is well-managed, secure, and deployment-ready. Using tools like Data Import Wizard and Data Loader, administrators can efficiently manage large and small data volumes. Duplicate Rules maintain data integrity by preventing redundant records, while Data Export & Backup safeguards against accidental data loss. Deployment tools such as Change Sets, Unmanaged/Managed Packages, ANT Migration Tool, and VS Code with SFDX provide flexible methods to move metadata and code between environments, enabling controlled development, testing, and production releases.
- Overall, this phase guarantees that AgriConnect's farmer, crop, order, and marketplace data remain accurate, secure, and synchronized across Salesforce orgs, supporting reliable operations and future scalability.

External Id	Name	Phone	Email	State	District
FRM-001	Rahul	9876543210	rahul@test.com	Telangana	Karimnagar
FRM-002	Anita	9876543211	anita@test.com	Andhra Pradesh	Guntur
FRM-003	Sunil	9876543212	sunil@test.com	Karnataka	Bangalore
FRM-004	Snetha	9876543213	snetha@test.com	Madhya Pradesh	Pune
FRM-005	Arjun	9876543214	arjun@test.com	Tamil Nadu	Chennai

